

User Manual

Sera - Low-Cost Offline Voice Assistant for Hospitality Services

1. System Overview

The Hotel Voice Assistant is an Android-based, fully offline room assistant. Guests speak requests naturally; the system recognizes speech on-device, classifies the intent, and routes the request to the correct hotel department with no internet connection required.

Components:

- **Guest App** - Android app installed on in-room tablet/phone
- **Staff Dashboard** - Browser-based dashboard for hotel staff
- **Backend Server** - Runs on hotel LAN (laptop or Raspberry Pi)

2. Guest Guide (Android App)

Getting Started

1. The device is pre-configured in your room; no setup needed.
2. Ensure the device is connected to the hotel WiFi.

Making a Voice Request

1. Tap the **microphone button** on screen.
2. Speak your request clearly (e.g., *"I need clean towels"*).
3. The app displays what it heard and the detected request.
4. A spoken confirmation is played automatically.
5. Your request is sent to the relevant department.

Tracking Your Request

- Tap **"View History"** to see all your submitted requests.
- Status updates: **Pending** → **In Progress** → **Completed**.
- You will receive a spoken notification when your request status changes.

Rating Service

- Once a request is marked **Completed**, a star rating prompt appears.
- Tap 1–5 stars to rate the service.

Cancelling a Request

- In your request history, tap a **Pending** request and select **Cancel**.

3. Supported Voice Commands

Category	Example Phrases
Food & Drinks	"I'd like to order breakfast", "Send me a bottle of water"
Room Cleaning	"Please clean my room", "I need housekeeping"
Towels	"I need fresh towels", "Can I get more towels?"
Toiletries	"I need shampoo and soap", "Send toothpaste please"
Pillow / Blanket	"I need an extra pillow", "Can I have another blanket?"
Laundry	"I need laundry service", "Please collect my laundry"
Maintenance	"The TV is not working", "My air conditioning is broken"
Temperature	"The room is too cold", "Can you adjust the temperature?"
Lighting	"The lights are not working", "Can you fix the lights?"
Wake-Up Call	"I need a wake-up call at 7am", "Wake me up at 6:30"
Taxi / Transport	"I need a taxi to the airport", "Call me a cab"
Checkout / Billing	"I'd like to check out", "Can I get my bill?"
Do Not Disturb	"Please do not disturb me"
Noise Complaint	"There is too much noise next door"
Concierge	"Can you recommend a restaurant?", "What are local attractions?"
Emergency	"I need emergency help"

Tip: Speak naturally - the system understands conversational English.

4. Staff Guide (Dashboard)

Accessing the Dashboard

Open a browser and go to: `http://<server-ip>:8000/dashboard`

Managing Requests

- All incoming guest requests appear in real time.
- Each card shows: Room number, request text, detected intent, and department.
- **Update Status:** Click a request → change status to *In Progress* or *Completed*.
- **Reassign Department:** Use the department dropdown if the auto-routing needs correction.

Messaging a Guest

- Open a request → click **Send Message** → type a message → Send.
- The guest receives the message as a spoken notification on their device.

5. IT Setup (Brief)

Backend

Install dependencies

```
pip install -r requirements.txt
```

Start the server

```
uvicorn app.main:app --host 0.0.0.0 --port 8000
```

- Runs on Python 3.9+, uses SQLite (no database configuration needed).
- Server must be on the same WiFi network as all guest devices.

Android App

1. Open android/ in Android Studio.
2. In `ServerConfig.kt`, set the backend server's local IP address.
3. Build and deploy to Android devices (Android 8.0+, min API 26).

Network Requirements

- Hotel LAN WiFi (all devices on same subnet).
- No internet connection required for STT or NLU - fully offline.

6. Troubleshooting

Issue	Fix
App cannot connect to server	Check WiFi connection; verify server IP in app settings
Voice not recognized	Speak clearly; reduce background noise; try again
Request sent to wrong department	Staff can reassign from the dashboard
App not responding	Restart the app; ensure backend server is running

For technical details, see the project repository README files.